🔐 ENTERPRISE NETWORK INTRUSION DETECTION & PREVENTION LAB

Using Snort

🛠 Tools & Technologies:

- Snort
- Kali Linux
- Ubuntu
- VirtualBox
- Nmap

🧑‍💻 Prepared By:

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📌 1. Introduction

Modern enterprise networks face continuous threats such as reconnaissance scanning, unauthorized access attempts, and malicious traffic injections.

To mitigate these risks, organizations deploy:

- **Intrusion Detection Systems (IDS)** – Detect and alert
- **Intrusion Prevention Systems (IPS)** – Detect and block

This project demonstrates the implementation of Snort as both IDS and IPS in a virtual lab environment.

🎯 2. Project Objectives

- Deploy Snort in IDS mode
- Detect ICMP (Ping) attacks
- Detect TCP SYN (Nmap) scans
- Configure Snort in IPS inline mode
- Block malicious traffic
- Perform log analysis

🌐 3. Lab Architecture

IDS Mode (Passive Monitoring)

Kali Linux (Attacker)

↓

Internal Network

↓

Ubuntu Server (Snort IDS)

Snort monitors traffic and generates alerts but does not block packets.

IPS Mode (Inline Blocking)

Kali Linux (Attacker)

↓

Ubuntu Server (Snort IPS Inline Mode)

↓

Target Network

Snort detects and blocks malicious packets in real time.

⚙️ 4. Installation & Setup

System Update:

```
sudo apt update
```

Install Snort:

```
sudo apt install snort -y
```

Verify Installation:

```
snort -V
```

📷 Screenshot 1 – Snort Installation Verification

```
tariqahmad@ubantu:~$ snort --v

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.15.1 GRE (Build 15125)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.10.1 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.2.11

tariqahmad@ubantu:~$
```

🧠 5. Custom Rule Configuration

Custom rules were added in:

```
/etc/snort/rules/local.rules
```

Ensure the following line exists in snort.conf:

```
include $RULE_PATH/local.rules
```

🔹 ICMP Detection Rule

```
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001;
rev:1;)
```

Purpose: Detect ping-based reconnaissance.

🔹 IPS Blocking Rule

```
drop icmp any any -> any any (msg:"ICMP Blocked"; sid:1000003; rev:1;)
```

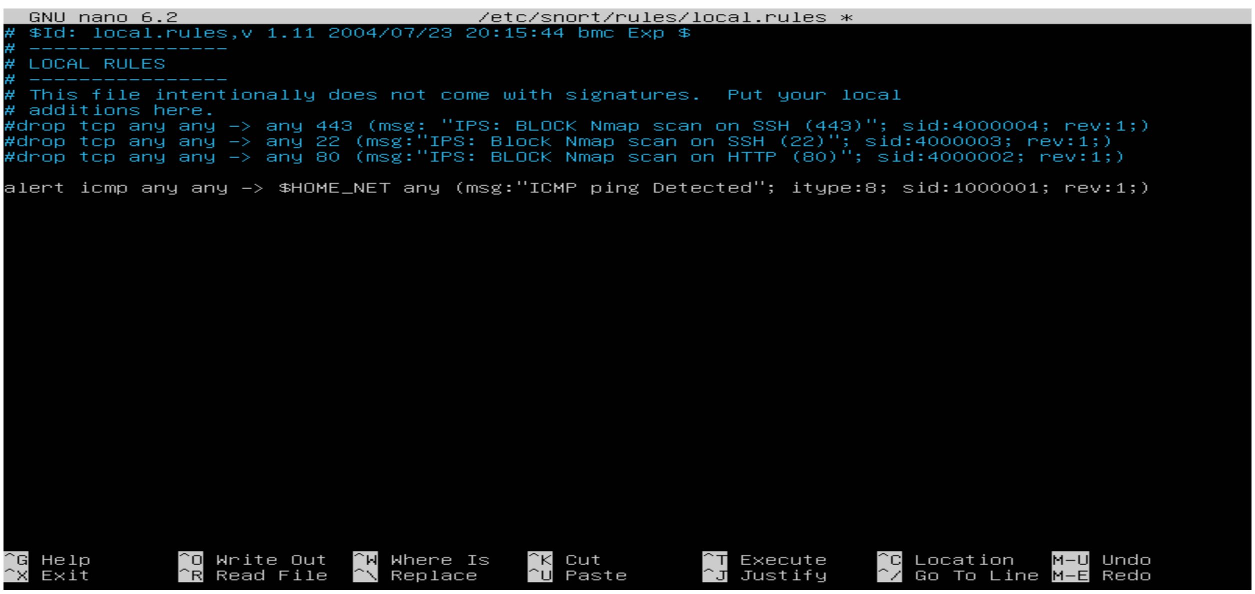
Purpose: Block malicious ICMP traffic.

🚨 6. Running Snort

IDS Mode

```
sudo snort -A console -c /etc/snort/snort.conf -i enp0s3
```

📸 Screenshot 2 – Snort Running in IDS Mode



Objective

The objective of this implementation was:

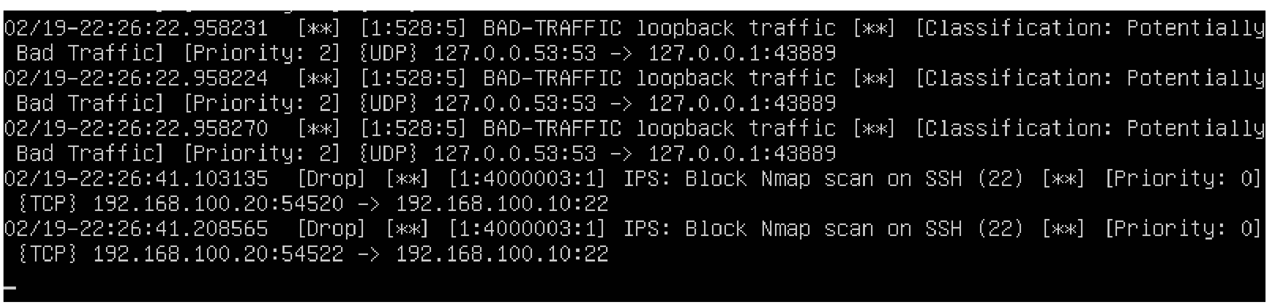
- To configure Snort in IPS mode
- To block SSH access attempts (Port 22)
- To block HTTP traffic (Port 80)
- To block HTTPS traffic (Port 443)

◆ Rule 1 – Block SSH (Port 22)

drop tcp any any -> any 22 (msg:"IPS: SSH Port 22 Access Blocked"; sid:1000101; rev:1;)

Purpose:

This rule blocks all TCP traffic attempting to access Port 22 (SSH).



```
02/19-22:26:22.958231  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:43889
02/19-22:26:22.958224  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:43889
02/19-22:26:22.958270  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:43889
02/19-22:26:41.103135  [Drop] [**] [1:4000003:1] IPS: Block Nmap scan on SSH (22) [**] [Priority: 0]
 {TCP} 192.168.100.20:54520 -> 192.168.100.10:22
02/19-22:26:41.208565  [Drop] [**] [1:4000003:1] IPS: Block Nmap scan on SSH (22) [**] [Priority: 0]
 {TCP} 192.168.100.20:54522 -> 192.168.100.10:22
```

◆ Rule 2 – Block HTTP (Port 80)

Purpose:

This rule blocks all HTTP web traffic on Port 80.

drop tcp any any -> any 80 (msg:"IPS: HTTP Port 80 Access Blocked"; sid:1000102; rev:1;)

```
02/19-09:58:45.436570  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:46231
02/19-09:58:45.436564  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:46231
02/19-09:58:45.436585  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:46231
02/19-09:58:47.903632  [Drop] [**] [1:4000002:1] IPS: BLOCK Nmap scan on HTTP (80) [**] [Priority: 0
] {TCP} 192.168.100.20:41573 -> 192.168.100.10:80
02/19-09:58:48.006258  [Drop] [**] [1:4000002:1] IPS: BLOCK Nmap scan on HTTP (80) [**] [Priority: 0
] {TCP} 192.168.100.20:41575 -> 192.168.100.10:80
_
```

◆ Rule 3 – Block HTTPS (Port 443)

Purpose:

This rule blocks secure HTTPS traffic on Port 443.

drop tcp any any -> any 443 (msg:"IPS: HTTPS Port 443 Access Blocked"; sid:1000103; rev:1;)

```
02/19-22:19:52.217703  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:38603
02/19-22:19:52.217696  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:38603
02/19-22:19:52.217723  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:38603
02/19-22:20:00.768792  [Drop] [**] [1:4000004:1] IPS: BLOCK Nmap scan on SSH (443) [**] [Priority: 0
] {TCP} 192.168.100.20:50280 -> 192.168.100.10:443
02/19-22:20:00.872470  [Drop] [**] [1:4000004:1] IPS: BLOCK Nmap scan on SSH (443) [**] [Priority: 0
] {TCP} 192.168.100.20:50282 -> 192.168.100.10:443
```

🔍 7. Attack Simulation

ICMP Ping Attack

Executed from Kali:

```
ping <target-ip>
```

Result:

- Detected in IDS mode

📸 Screenshot 3 – ICMP Ping Detected

```
02/18-23:33:29.284768  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:51640
02/18-23:33:29.284804  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:51640
02/18-23:33:29.285077  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.1:34428 -> 127.0.0.53:53
02/18-23:33:29.285164  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.1:34428 -> 127.0.0.53:53
02/18-23:33:29.285155  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.1:34428 -> 127.0.0.53:53
02/18-23:33:29.285185  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.1:34428 -> 127.0.0.53:53
02/18-23:33:29.285374  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:34428
02/18-23:33:29.285406  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:34428
02/18-23:33:29.285400  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:34428
02/18-23:33:29.285421  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:34428
02/18-23:33:29.285563  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.1:34428 -> 127.0.0.53:53
02/18-23:33:29.285613  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.1:34428 -> 127.0.0.53:53
02/18-23:33:29.285605  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.1:34428 -> 127.0.0.53:53
02/18-23:33:29.285630  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.1:34428 -> 127.0.0.53:53
02/18-23:33:29.325338  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:34428
02/18-23:33:29.325376  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:34428
02/18-23:33:29.325368  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:34428
02/18-23:33:29.325389  [**] [1:528:5] BAD-TRAFFIC loopback traffic [**] [Classification: Potentially
 Bad Traffic] [Priority: 2] {UDP} 127.0.0.53:53 -> 127.0.0.1:34428
```

8. IPS Mode (Blocking Demonstration)

Snort was executed in inline mode:

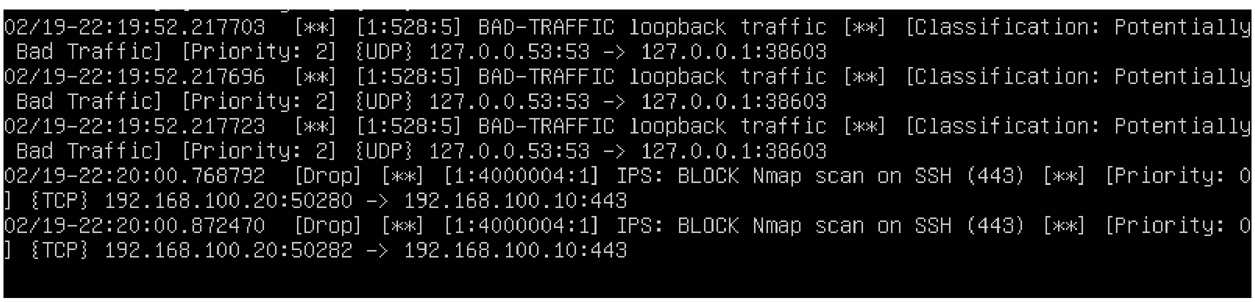
```
sudo snort -Q --daq afpacket -c /etc/snort/snort.conf -i eth0
```

Ping attack was repeated.

Result:

- ICMP traffic was blocked.

Screenshot 5 – ICMP Blocked in IPS Mode



📈 10. Results

✓ Successful ICMP detection

✓ Successful TCP SYN detection

✓ Successful ICMP blocking in IPS mode

✓ Proper logging and alert generation

🏁 11. Conclusion

This project demonstrates the successful deployment of Snort as both an Intrusion Detection and Intrusion Prevention system in a controlled lab environment.

The system effectively detected reconnaissance attacks and blocked malicious traffic, simulating real-world SOC operations.

This hands-on implementation strengthened knowledge in:

- Network Security
- IDS/IPS Deployment
- Linux Administration
- Attack Simulation

🚀 12. Future Enhancements

- Integration with SIEM tools
- Email alert notifications
- ELK Stack visualization

- Brute-force detection rules
- Web attack detection

🔐 Final Statement

This project reflects practical cybersecurity defense implementation and readiness for entry-level SOC Analyst roles.